The Same 2k Pool at Four Epochs

Status: Owner training/evaluation completed; final reported local candidate at project closeout. Date: 2026-10-17.

Question and Controls

The owner requested a fresh four-epoch run on the same expanded 2k dataset. Compare against that dataset's two-epoch run, not the earlier 1k duration trial. The unweighted compact/name-first checkpoint remains the historical development reference. The four-epoch candidate's measured result is recorded below; no statistical significance or untouched-test generalization is established.

Only `num_train_epochs` changes from 2 to 4. Run/config identity, output paths and config hash also change so the two-epoch artifact is not overwritten. Data, split membership, prompt, LoRA, learning rate, seed, loss weighting and laptop overrides stay fixed. This starts from the base model with a fresh LoRA adapter, not from the saved two-epoch adapter. The cosine schedule spans the longer training horizon, so this is not an identical first two epochs followed by extra updates.

Identity

Item	Value
Config	`configs/sft/qwen2_5_1_5b_lora_coverage_2k_v2_epochs4.json`
Config SHA256	`da38eb64730f8db30b1bf1a5d6f5296a805351598ca2057a1f70e6874141f2e5`
Dataset	`data/sft/v1/name-first-coverage-2k-v2`
Dataset revision	`43892841fad91d9c328f3577e831c5d192242f84ceabe82d18e7deb30d2c6ab4`
Base model	`Qwen/Qwen2.5-1.5B-Instruct`
LoRA	r=8, alpha=16, Q/K/V/O, dropout=0.05
Learning rate / epochs / seed	2e-4 / 4 / 1
First-tool loss weighting	1x, unweighted
Laptop settings	Context 1,024, batch one, accumulation 16
Output	`models/qwen2_5_1_5b_lora_coverage_2k_v2_epochs4/adapter`

The owner reported 1,952 usable training rows from a 2,000-row raw cap: 800 BAYMAX, 1,152 xLAM, 200 clarification and 200 refusal rows, with zero exact duplicates. The same validation cap yields 195 usable rows. These are existing artifact counts, not a new four-epoch measurement. At the same usable counts, four epochs

give approximately twice the sample exposure of two epochs. Model quality is measured separately by the saved evaluation below.

Owner Command

```
.venv\Scripts\python scripts\train_sft_lora.py `
  --config configs\sft\qwen2_5_1_5b_lora_coverage_2k_v2_epochs4.json `
  --local-low-vram `
  --max-train-rows 2000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

No dataset rebuild or additional examples are needed. The coding agent did not run training or evaluation. Preserve raw results and compare both durations under the same compact JSON input, greedy decoding, 150-scenario suite and scoring. Report task success, routing macro-F1, invalid outputs and the decision matrix; lower training loss alone is not a model-quality win. This repeatedly inspected suite is development evidence, not an untouched final test.

Saved Evaluation Results

Sources: two epochs and four epochs. The four-epoch result is timestamped 2026-10-17T16:52:19Z. Both use the same 150 scenarios, compact JSON, temperature zero, at most 256 generated tokens and fake adapters. Values below are transcribed from saved artifacts, not re-evaluated.

Measurement	Two epochs	Four epochs
Task successes / 150	64	73
Task success rate	0.427	0.487
Routing macro-F1	0.615	0.681
Invalid outputs / 150	2	0
Tool-call accuracy	0.667	0.753
Argument accuracy	0.806	0.836
Clarification wording accuracy	0.163	0.350
Refusal wording accuracy	0.603	0.690
Conditional decision accuracy	0.649	0.707
Decision coverage	0.987	1.000
Hallucination rate	0.000	0.007
Mean inference latency, ms	2,758	5,652
Reported API cost, USD	0.0000	0.0000

Four epochs, expected rows and predicted columns:

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	62	13	6	0
CLARIFY	17	17	6	0
REFUSE	1	1	27	0

The recorded four-epoch run completes nine additional tasks and has better macro-F1 than the two-epoch counterpart. Compared with the historical unweighted 1k reference, it completes 73 versus 57 tasks and records macro-F1 0.681 versus 0.612. This is a useful development result for closing the project, not proof that four epochs always help: the earlier 1k four-epoch trial regressed, and no repeated training seeds were measured here.

Clarification remains incomplete: 17/40 requests that should clarify still cause ACT, and 6/40 cause REFUSE. ACT recall is 62/81, and refusal recall is 27/29. The recorded latency is higher; hardware/background conditions were not controlled well enough to attribute that solely to the training duration. A zero local API cost field does not mean zero hardware, energy or training cost.

The project owner accepted the work as complete on 2026-10-17. This does not imply production readiness, real integrations, a new MLX checkpoint release or an untouched final evaluation. Model weights and generated training data remain local and are not part of the Git closeout.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.